Single-target batch floor in the declared m8 model

November 13, 2026. Unsigned producer evidence [pcc:unverified].

Scope. This corrects one incoming entropy-charge cost projection. It is not a lower bound on an actual sparse solver or ECDLP, and it does not close PKM or another encoding.

Let H_2 denote binary entropy and $\theta_8 = 8H_2(1/8)$. Exact Lean bounds give

$$\theta_8 = 24 - 7\frac{\log 7}{\log 2}, \qquad 4 < \theta_8 < 5.$$

The incoming unclamped exponent and its balanced whole-attack call exponent obey

$$\frac{2}{5} < \frac{2}{9-\theta_8} < \frac{1}{2}, \qquad 1 - \frac{7}{9-\theta_8} < 0.$$

The apparent gain is paired with fewer than one total solver call. Even allowing all useful outputs per call, enforce the optimistic single-target batch floor:

$$W(p,b,c) = \max\{1, p/b^7\}\, c + b^2, \quad p > 1,\ b \geq 1,\ c \geq b^4.$$

The declared charge $c = b^{\theta_8}$ meets that bound. Lean establishes

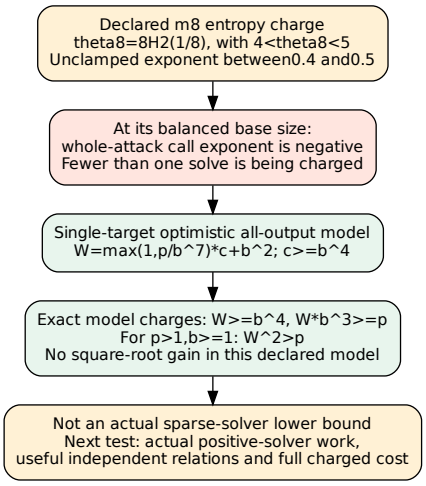
$$p < W(p,b,b^{\theta_8})^2 \quad \text{for every } p > 1, b \geq 1.$$

No wall time, GAE, measured solver exponent or universal cryptographic bound is asserted.

Proof. For charged work t, the model implies $b^4 \leq t$ and $p \leq tb^3$. If $b > 1$, then $p < pb \leq tb^4 \leq t^2$. If $b = 1$, then $t \geq p > 1$, so $t^2 > p$. The central surplus identity is

$$(tb^3 - p)b + (t - b^4)t = t^2 - pb.$$

The entropy bounds use the exact inequalities $2^{19} < 7^7 < 2^{20}$.



Evidence scope and the corrected next test

What the review changes. The D8,3 cell is still a degree diagnostic; its result alone does not decide the advertised single-target gain under this charge model. Actual sparse solver work may be lower than the dense proxy. Measure positive-solver work, witness extraction, actual solve count and useful independent relations before selecting an expensive new run. A fixed exponent below1/2 is still exponential in input bits; the original objective requires a uniform growing-parameter bit-cost bound.

Bounded source observations. Incoming snapshot `5965fa03b`. L1's generator explicitly enforces rational targets; its hash matches the manifest. Its level9 closure is censored; an F4 degree10 trace is distinct from the frozen completed-closure condition. The L1 manifest names closure Rust files absent from the inspected committed study listing; replay-source recovery remains open. L2 itself limits truncated ideal searches. L3's checker distinguishes unsplit degrees from alternate encodings and full-space size from actual solver cost. L4's checker flags success conditioning and mask/filter scope. L5's point-ordered widths do not exclude arbitrary algebraic/interleaved representations. These are source observations, not independent numerical replays or a blanket closure. Full original protocol and code review of all five leads is unfinished.

Formal map. `PrimeICCostFloor.lean`, frozen source `d42ef1874`. Twelve declarations cover the universal surplus, explicit charge inequalities, actual defined work expression, exact entropy formula and bounds, unclamped exponents and corrected entropy work. Exact hypotheses and declaration bindings are linked from the Markdown report. All producer PCC standing is unsigned.

Checks. Lean4.35.0 with unchanged pinned dependencies. Build003 passes2074 jobs. Full tracked-source audit155 declarations, zero failing and zero extra trusted axioms; fresh import-inclusive completion has its own receipt. Standard foundations only. Build001 had power-rewrite/positivity proof issues; build002 corrects them and build003 adds exact exponent bounds. Failed and completed logs are retained; earlier source hashes are unchanged.

Native record `IDC1h0f32be38dee24607` checks only the universal surplus identity at32+32 points using the unchanged checker and exact universal source. It does not execute Lean or prove inequalities, logarithms or actual solver cost. Its adapter's22 inherited tests pass. No runtime claim.

Next protocol. Recover pinned native source; separate the degree cell from actual relation finding on online masked targets. Charge at least one whole solve, base generation, rejected candidates, useful independent output, linear algebra and final scalar verification. Retain controls/censoring and isolate hosts for time comparisons. Kummer eligibility $2^t \mid p-1$ and all symmetry/bit costs must be explicit. The draft is not executed: isolab monthly budget is exhausted. No new paid job, benchmark, leaderboard or ecdb edit.

Sources. Exact incoming L1 protocol/amendments/verdict/generator; L2 report, L3/L4 checker reports and L5 report; Galbraith15.8.1–2 from the previous source-pinned author-hosted book; pinned logarithm/power/inequality APIs. KB/conductor unavailable, bounded fallback; no priority claim for elementary batch accounting. Editable DOT/SVG and source-linked report/evidence accompany this PDF.